

22GE004-BASICS OF ELECTRONICS ENGINEERING

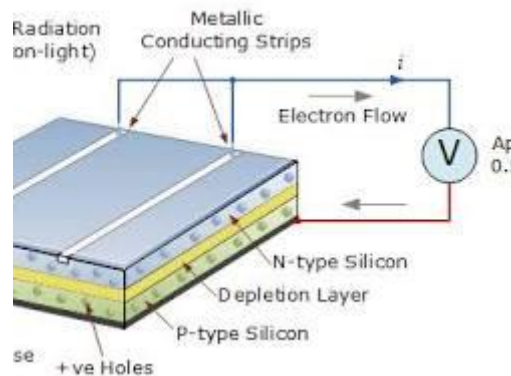
UNIT V & DEVICES FOR SPECIAL REQUIREMENTS

Light to Electrical Energy Conversion Using Solar Cells

Introduction

Solar cells, also known as photovoltaic cells, are devices that convert sunlight directly into electricity. This¹ process, called photovoltaic effect, has gained significant attention as a renewable energy source. Solar cells have become increasingly popular due to their environmental friendliness, low operating costs, and potential to generate clean energy.

Working Principle of Solar Cells



solar cell structure with photons striking the pn junction

A solar cell typically consists of a semiconductor material, usually silicon, with a p-n junction. When sunlight strikes the solar cell, photons in the sunlight excite electrons in the semiconductor material, creating electron-hole pairs.

- **Photon Absorption:** Photons from sunlight are absorbed by the semiconductor material.
- **Electron-Hole Pair Generation:** The absorbed energy excites electrons, freeing them from their bonds and creating electron-hole pairs.



- **Charge Separation:** The p-n junction creates an electric field that separates the electrons and holes.
- **Current Generation:** The separated charges flow through an external circuit, generating electric current.

Types of Solar Cells

1. **Crystalline Silicon Solar Cells:**
 - **Monocrystalline Solar Cells:** Highly efficient but more expensive.
 - **Polycrystalline Solar Cells:** Less efficient but more cost-effective.
2. **Thin-Film Solar Cells:**
 - **Amorphous Silicon Solar Cells:** Flexible and lightweight but lower efficiency.
 - **Cadmium Telluride (CdTe) Solar Cells:** High efficiency and good performance in low-light conditions.
 - **Copper Indium Gallium Selenide (CIGS) Solar Cells:** High efficiency and good performance in low-light conditions.

Advantages of Solar Energy

- **Renewable Energy Source:** Solar energy is a clean and renewable source of energy, reducing dependence on fossil fuels.
- **Reduced Carbon Footprint:** Solar energy helps reduce greenhouse gas emissions and mitigate climate change.
- **Lower Operating Costs:** Once installed, solar energy systems have minimal operating costs, making them a cost-effective long-term solution.
- **Energy Independence:** Solar energy can reduce reliance on traditional energy sources and promote energy independence.
- **Versatile Applications:** Solar energy can be used for residential, commercial, and utility-scale power generation.

Challenges and Future Trends

- **Initial Cost:** The initial cost of solar energy systems can be high.
- **Intermittent Nature of Solar Energy:** Solar energy generation depends on sunlight availability, which can be affected by weather conditions.
- **Land Requirement:** Large-scale solar power plants require significant land area.

Despite these challenges, advancements in solar cell technology and government incentives are driving the growth of solar energy. Future trends include:

- **Perovskite Solar Cells:** High-efficiency and low-cost solar cells with potential for further improvement.
- **Solar Cell Integration with Building Materials:** Integrating solar cells into building materials to create energy-efficient structures.
- **Solar Energy Storage:** Developing advanced battery storage technologies to store excess solar energy for use during nighttime or cloudy days.



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

An Autonomous Institution Affiliated to Anna University Chennai - Approved by AICTE - Accredited by NAAC with "A+" Grade

SATHYAMANGALAM - 638 401 ERODE DISTRICT TAMIL NADU INDIA

Ph: 04295-226000 / 221289 Fax: 04295-226666 E-mail: stayahead@bitsathy.ac.in Web: www.bitsathy.ac.in

Summary

Solar energy is a promising renewable energy source with the potential to significantly reduce our reliance on fossil fuels. As technology continues to advance, solar cells are becoming more efficient, affordable, and accessible. By harnessing the power of the sun, we can contribute to a sustainable and cleaner future.

DISCUSSION QUESTIONS:

- 1. Identify the usage of Solar cell.**
- 2. Indicate the applications of Solar cell**
- 3. Infer the advantages of Solar cell.**